

The Hermitian Cope

*An open letter on the physics community's
centuries-old anxiety about i*

C. Paci — March 2026

Dear colleagues,

We need to talk about the Hermitian inner product.

Not about whether it works — it works beautifully, with the mechanical efficiency of a machine designed to destroy evidence. I want to talk about *why* you chose it, and what that choice has cost us.

The Conjugation Ritual

Every quantum mechanics textbook performs the same ritual in chapter two. We define a state $|\psi\rangle$ that lives in a complex Hilbert space, granting it the full richness of amplitude *and* phase. Two real degrees of freedom per component. The complete description. Then, five pages later, we define the inner product:

$$\langle\phi|\psi\rangle = \sum_n \phi_n^* \psi_n$$

and in that single conjugation — that silent little star — we annihilate exactly the thing that made the space complex in the first place.

This is not a mathematical necessity. It is a *choice*. A choice motivated not by physics, but by fear.

What Are You Afraid Of?

The fear has a name. It is the fear of *complex observables*. The Hermitian inner product guarantees $\langle\psi|\hat{A}|\psi\rangle \in \mathbb{R}$ for self-adjoint $\hat{A}$, and the entire interpretive apparatus of quantum mechanics is built on this guarantee. Eigenvalues are real. Probabilities are real. Expectation values are real. Everything that touches experiment is scrubbed clean of i before it reaches a human mind.

And why? Because we decided, a priori, that measurement outcomes *must* be real numbers. Not because nature told us so. Because Hermite and von Neumann found it aesthetically comfortable. Because the alternative — that a measurement might return a complex number, that i might have operational meaning at the level of observation — was too disturbing to entertain.

So we built a formalism that *structurally prevents* us from ever encountering the imaginary part. Then we celebrated the formalism’s “consistency” as though we had discovered something, rather than enforced it.

The Born Rule: Arson as Epistemology

It gets worse. Having conjugated away the relative phase in the inner product, we then apply the Born rule:

$$P = |\psi|^2 = \psi^* \psi$$

This is not a derivation. It is a *policy*. The policy states: take the most informationally rich object in physics — a complex amplitude carrying both magnitude and phase — and *square it into a real number*. Throw away the phase. Keep only the shadow.

Then, having burned the letter, we complain that we cannot read it.

The measurement problem is not a mystery. It is a *consequence*. You chose an inner product that deletes phase information at the moment of observation, and now you are confused about where the information went. You chose $|\psi|^2$ over ψ , reality over complexity, comfort over completeness. The measurement problem is the invoice for that choice, and it has been sitting unpaid on your desk for a hundred years.

The Historical Anxiety

We should not be surprised. The physics community’s relationship with $\sqrt{-1}$ has always been one of repressed panic.

Cardano called complex numbers “as subtle as they are useless.” Descartes coined the term *imaginary* — a word designed to delegitimize. Euler domesticated them with $e^{i\theta}$, and even then, the consensus was that complex numbers were a *trick*: a computational shortcut that should never appear in a final answer. Real answers for real physics. The complex machinery hums away backstage, but the audience must never see it.

The Hermitian inner product is the 20th-century version of this same anxiety. We let i into

the state space because we had no choice — Schrödinger’s equation demands it. But we immediately installed a conjugation gate at every exit. No complex number gets out alive. The formalism uses i everywhere and trusts it nowhere.

What i Actually Does

Let us be honest about what the imaginary unit contributes to quantum mechanics, and then ask whether we should be so eager to conjugate it away.

Phase is what makes quantum mechanics quantum. Without phase, there is no interference. Without interference, there is no double-slit experiment, no tunneling, no chemistry, no entanglement. The entire non-classical structure of the theory lives in the relative phases between amplitudes.

And phase is *exactly* what the Hermitian inner product destroys.

The Berry phase says: “phase has geometric meaning.” The Aharonov-Bohm effect says: “phase has physical consequences even where fields vanish.” Interference says: “phase determines which outcomes occur and which are forbidden.” Nature is screaming at us that phase is not a bookkeeping artifact. Phase is doing the work.

And our response? Conjugate it away. Square the amplitude. Report the real part. Move on.

The $\mathcal{PT}$ -Symmetric Hint

Carl Bender showed two decades ago that $\mathcal{PT}$ -symmetric Hamiltonians — non-Hermitian, openly complex — can have entirely real spectra. You do not *need* Hermiticity for real eigenvalues. You need a weaker condition. The physics community responded with polite interest and continued teaching the Hermitian axiom as though Bender had not spoken.

This is revealing. When someone demonstrates that your axiom is *sufficient but not necessary*, and you continue treating it as necessary, you are no longer doing mathematics. You are doing theology.

The Ternary Observation

From the framework I work in, the diagnosis is specific. A cubic lattice with ternary states $\{-1, 0, +1\}$ naturally generates dynamics through $e^{i\theta}$, where θ is not a gauge artifact but the *physical phase* of the field. The update rules produce interference. The interference produces

structure. The structure produces everything we observe.

At no point does the formalism require us to conjugate away the phase. The phase is not a nuisance to be managed. It is the mechanism by which void becomes matter.

When you take $|\psi|^2$, you are photographing a dance and calling the photograph the dancer. You have captured the *intensity* and lost the *motion*. The measurement problem is the question: “Why does the dancer stop moving in photographs?” She doesn’t. You chose a camera that only records brightness.

The Invitation

I am not suggesting we abandon the Hermitian inner product. It works. Predictions match experiment. The machinery is powerful.

I am suggesting we recognize it for what it is: a *coping mechanism*. A centuries-old strategy for making the imaginary unit palatable by ensuring it never appears in a final answer. A formalism that *systematically prevents* us from asking whether i has observational content, and then interprets the resulting silence as evidence that it doesn’t.

The measurement problem, the collapse postulate, the interpretation wars, the retreat into “shut up and calculate” — these are not separate mysteries. They are symptoms. Symptoms of a theory that was given the full complex plane and immediately flinched.

i is not imaginary. It is not a trick. It is not a convenience.

It is the part of the answer you have been conjugating away for a hundred years.

Perhaps it is time to stop.

— C.P.

*Disclaimer: This letter represents a theoretical perspective within Foundational Ternary Dynamics (FTD). The Hermitian inner product remains the standard and empirically successful framework. The author’s quarrel is with the **interpretive assumption** that phase must be eliminated at observation, not with the formalism’s predictive power.*